



PHYSICS 3

ACADEMIC

Grade 11/12

Curriculum Guide

PHYSICS 3
Pre-requisites: Physics 1, Physics 2

Course Description:

Physics 3 builds on the concepts and scientific skills developed in Physics 2. The course focuses on the principles of electricity and electromagnetism, including electric charges, electric fields, electric circuits, magnetism, and electromagnetic induction. The learners develop conceptual understanding and quantitative problem-solving skills as they investigate the relationships among electrical quantities and the interactions between electricity and magnetism.

Through experiments, modeling, and real-world applications, the learners apply physical principles to explain the operation of electrical devices and technologies such as motors, generators, and household electrical systems. The course also highlights the role of electricity and electromagnetism in modern technologies and emphasizes safe and responsible use of electrical systems.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Electricity		
1. Electrostatics	1. the study of electrostatics is essential in explaining the electrical nature of matter; and	1. explain the nature of charge and different charging processes;
		2. calculate the total electrostatic force on a point charge due to a system of charges in one-dimensional or two-dimensional spaces;
		3. describe the nature of the electric field due to a single point charge;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
		4. determine the magnitude and direction of the electrostatic force on a static point charge in a region of a constant electric field; 5. explain the factors affecting electric potential and its distinction from electric potential energy;
2. Current Electricity	2. the relationships among electrical quantities, such as electric potential difference (voltage), current, and resistance are important in maintaining efficient and safe electrical systems;	6. design and conduct an experiment to graphically validate the relationship among current, resistance, and voltage in direct-current circuits (series, parallel, and combination) as stated in Ohm's Law; 7. explain how the geometric properties of a capacitor, particularly plate area and plate separation, affect capacitance; 8. differentiate series and parallel capacitor configurations and describe their effects on the circuit's total capacitance; and 9. solve problems involving Ohm's Law, resistance, and capacitance in a circuit.
Performance Standard The learners can demonstrate their understanding of the principles of electrostatics and current electricity by using theoretical and experimental approaches to solve context-rich problems and situations involving these branches of Physics.		
Suggested Performance Task Design and construct a miniature functional direct current (DC) electrical circuit for a model room that incorporates switches and lighting while applying safe wiring practices.		
Electromagnetism		
3. Magnetism	3. magnetism is a property of certain materials arising from the	10. describe how the alignment of magnetic domains in materials produces magnetism and how this relates to magnetic and non-magnetic materials;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
	alignment of atomic magnetic domains, and that natural magnetic phenomena such as Earth's magnetic field influence navigation and living organisms;	11. explain the characteristics and significance of Earth's magnetic field, including its role in navigation and its uses and effects on living organisms;
4. Electromagnetism	4. a moving electric charge changes the characteristics of the space around it by establishing a magnetic field in addition to its electric field, enabling moving charges to interact through magnetic forces; and	12. calculate the magnetic force acting on a moving electric charge and a current-carrying conductor in a uniform magnetic field;
		13. interpret diagrams and relationships describing the magnetic fields produced by current-carrying conductors and solenoids.;
		14. explain how different factors affect the magnitude of an induced electromotive force based on Faraday's law of electromagnetic induction;
		15. interpret Lenz's Law because of the conservation of energy in electromagnetic induction;
		16. analyze context-rich problems using Faraday's law of electromagnetic induction;
		17. explain the difference between direct current (DC) and alternating current (AC) and their applications;
	5. the study of electromagnetism explains the relationship between electric and magnetic fields, electromagnetic induction, and the technologies that make modern life possible.	18. explain the working principles of electric motors, electric generators, and transformers using electromagnetic induction; and 19. describe how electromagnetic induction is applied in technologies such as maglev trains, metal detectors, traffic sensors, and magnetic cards.

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Performance Standard The learners can explain the principles of electromagnetism through theoretical and experimental approaches and can recognize their significance in electrical devices.		
Suggested Performance Task Develop an electromagnetism kit that allows users to build and explore various electromagnetic devices such as an electric motor and a hand-crank generator to power LEDs or small devices.		